

Yehoshua ("Shua") Halle

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EDUCATION

Carnegie Mellon University (CMU) Pittsburgh, PA

Anticipated Graduation May 2027

Master of Robotics Systems Development | GPA: 3.67/4.00

Relevant courses: Autonomous Vehicles; Robot Autonomy; Robot Mobility; Computer Vision; Systems Engineering;

University of Maryland (UMD) College Park, MD

Graduated with Honors December 2024

Bachelor of Mechanical Engineering | GPA: 3.84/4.00

Minor of Robotics and Autonomous Systems

WORK EXPERIENCE

Program Director of Computer Science, TIC Summer Camp (Bethesda, MD)

July 2025 - Aug 2025

- Managed a team of 9 counselors to teach programming, robotics, and web concepts to kids of ages 6-15
- Developed new training material about teaching technical subjects to campers. The CEO shared it with all other camp sites due to its positive impact.
- Resolved unpredictable situations including staff conflicts, parental communication, and camper misbehavior
- Adapted activities, rules, equipment and staff duties as necessary to meet needs of campers
- Facilitated collaborative problem-solving with staff through daily open-ended discussions

Associate Researcher, National Institute of Standards and Technology (Gaithersburg, MD)

May 2024 - Aug 2024

- Developed a synchronized digital model of a UR5e robot arm using ROS2 for real-time control and monitoring
- Initiated independent learning of ROS2 and C++
- Evaluated effectiveness of the ISO Standard for Digital Twins in Manufacturing by strictly following its procedures for and documenting its positive or negative contribution towards implementation
- Communicated concepts and results from case study in a presentation and draft research letter

PROJECTS

Powertrain Team Leader, Terps Racing Baja SAE Team (College Park, MD)

Jan 2023 - Aug 2024

- Coordinated a team of 10 students to design and manufacture a custom powertrain for an off-road vehicle
- Managed student design projects with GANTT chart, weekly meetings, and design reviews
- Integrated powertrain with other subsystems by checking for part collisions, verifying assembly operations, performing load calculations, and enforcing competition rules compliance
- Optimized transmission using C++ simulation and physical trials to improve 150 ft acceleration time by 1.6s: from 5.98s (top 25%) to 4.34s (top 15%)

"Assistive Feeder" Project, Volunteers for Medical Engineering (College Park, MD)

Jan 2023 - May 2024

- Designed a robot to help people with disabilities eat meals with more independence at an affordable price
- Programmed kinematics solver for Arduino microcontroller in C++
- Budgeted bill of materials to keep cost of device below \$250 target by minimizing shipping costs and using commodity electronics parts
- Tested prototype with potential end users to evaluate effectiveness and user experience
- Reviewed designs with stakeholders to ensure product requirements were met

SKILLS

Software: ROS/ROS 2, MATLAB, Simulink, SOLIDWORKS, Inventor, Fusion 360, Linux OS, Microsoft Office

Programming Languages: Python, C, C++, CMake, Java, Lua, Bash

Engineering: CAD, CAM, GD&T, FEA, DFM, DFA, Computer Simulation, 3D Printing

Math: Controls Theory, Optimization, Numerical Methods, Statistics, Linear Algebra, Differential Equations